

Sentiment-Driven Crypto Trader Analysis

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Project Objective

The objective of this project was to analyze the relationship between Bitcoin market sentiment and trader performance using historical trading data and the Fear & Greed Index. The analysis focused on understanding how sentiment influences profitability, win rates, trading activity, and trader behavior.

Datasets Used

Historical Trader Data

- 211,224 trading records
- Information including account, execution price, trade size, side, and closed PnL

Bitcoin Fear & Greed Index

- 2,644 daily sentiment records
- Categories:
 - Extreme Fear
 - Fear
 - Neutral
 - Greed
 - Extreme Greed

Key Findings

Finding 1: Extreme Greed Produced the Best Performance

Extreme Greed periods generated the highest average profitability and the highest win rate among all sentiment categories.

- Average PnL: **67.89**
- Win Rate: **46.49%**

This suggests that traders generally perform better during strong bullish market sentiment.

Finding 2: Fear Periods Generated the Highest Trading Activity

Fear periods recorded the largest number of trades across all sentiment categories.

This indicates that traders become significantly more active during uncertain and volatile market conditions.

Finding 3: Infrequent Traders Outperformed Frequent Traders

Analysis showed that infrequent traders achieved higher average profitability compared to frequent traders.

This suggests that selective and disciplined trading may outperform excessive trading activity.

Finding 4: High Win Rate Does Not Always Mean Higher Profitability

The Consistent Winner segment did not generate the highest average PnL.

This indicates that profitability depends not only on winning frequency but also on position sizing, trade quality, and risk management.

Strategy Recommendations

Strategy 1: Sentiment-Aware Trading

Increase participation during Extreme Greed conditions while maintaining strict risk controls and profit-taking rules.

Strategy 2: Selective Trading During Fear

Instead of increasing trade frequency during Fear periods, focus on high-conviction opportunities and avoid overtrading.

Conclusion

The analysis demonstrates a meaningful relationship between market sentiment and trader behavior. Extreme Greed periods delivered the strongest trader performance, while Fear periods generated the highest trading activity. The findings suggest that sentiment indicators can be incorporated into trading and risk management frameworks to improve decision-making and overall trading outcomes.